SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:40**

Code of Conduct:

I __Avishek N (192511009)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Avishek N

INTRODUCTION

Knowledge Representation and Reasoning is a fundamental area of Artificial Intelligence concerned with representing information in a form that an intelligent agent can use to derive conclusions. Propositional Logic represents facts as propositions and connects them using logical operators, while First-Order Logic extends this representation using predicates, variables and quantifiers. Inference techniques such as Forward Chaining, Backward Chaining, Unification and Resolution allow an agent to derive valid conclusions from a knowledge base.

This case study applies these concepts to four practical AI scenarios: a smart medical diagnosis system, an autonomous traffic management agent, an agricultural expert reasoning system and a Wumpus World knowledge agent. Each case is solved using the exact rules and facts supplied in the assessment question. Where the supplied rule set is insufficient to establish a conclusion, that limitation is explicitly identified rather than assuming an unstated rule.

The uploaded assessment identifies CO3 as analysis of knowledge representation and reasoning using propositional logic, FOL, inference, resolution and chaining methods. □filecite□turn0file0□L5-L10□

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

The case gives three diagnostic rules and the facts Fever=True, Cough=True and Breathlessness=True for Patient A. □filecite□turn0file0□L18-L25□

(a) Propositional Logic Representation

Symbol	Meaning
F	Fever
C	Cough
R	Rash
B	Breathlessness
Flu	Possible Flu
Measles	Possible Measles
P	Possible Pneumonia
$\land$	AND
$\rightarrow$	Implies

Rule 1: $F \land C \rightarrow Flu$

Rule 2: $F \land R \rightarrow Measles$

Rule 3: $C \land B \rightarrow P$

Facts: F, C, B are True.

Knowledge Base: $KB = \{(F \land C \rightarrow Flu), (F \land R \rightarrow Measles), (C \land B \rightarrow P), F, C, B\}$.

(b) Forward Chaining

Forward Chaining begins with known facts and fires every rule whose antecedent is satisfied.

Step 0 – Initial facts: F, C, B.

Step 1 – Rule 1: $F \land C$ is true, therefore Flu is derived.

Step 2 – Rule 2: F is true but R is not known; therefore Measles is not derived.

Step 3 – Rule 3: C ∧ B is true, therefore P (Pneumonia) is derived.

Final: Possible Flu and Possible Pneumonia are derivable; Possible Measles is not derivable.

Step	Rule	Condition	Result
0	Initial	F,C,B	Known facts
1	R1	F∧C	Flu
2	R2	F∧R	Not fired
3	R3	C∧B	Pneumonia

Forward-Chaining Flowchart

Mermaid-style logic flow:

START → Load F,C,B → Check F∧C → YES: Derive Flu → Check F∧R → NO: No Measles → Check C∧B → YES: Derive Pneumonia → END

(c) Backward Chaining for Pneumonia

Goal: P (Pneumonia).

Supporting rule: C ∧ B → P.

Goal reduction: Prove C and prove B.

C: Cough is a given fact → TRUE.

B: Breathlessness is a given fact → TRUE.

Conclusion: Both subgoals are satisfied; therefore P is proven.

Goal tree:
Pneumonia (P)
├── Cough (C) → TRUE
└── Breathlessness (B) → TRUE
∴ Pneumonia → TRUE

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

The supplied FOL rules connect emergency vehicles with clear paths, clear paths with green signals, and vehicles behind a green-signal vehicle with permission to proceed. filecite turn0file0 L32-L38

(a) Unification and MGU

Rule 1: ∀x (Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)). Facts include Vehicle(Ambulance) and EmergencyType(Ambulance).

Match 1: Vehicle(x) with Vehicle(Ambulance) gives θ_1={x/Ambulance}.

Match 2: EmergencyType(x) with EmergencyType(Ambulance) gives θ_2={x/Ambulance}.

Combined MGU: θ={x/Ambulance}.

Conclusion: ClearPath(Ambulance).

Expression	Fact	Substitution
Vehicle(x)	Vehicle(Ambulance)	{x/Ambulance}
EmergencyType(x)	EmergencyType(Ambulance)	{x/Ambulance}

ClearPath(x)	ClearPath(Ambulance)	θ={x/Ambulance}

(b) Resolution Proof of Proceed(CarA)

Relevant CNF clauses:

C1: ¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

C2: ¬ClearPath(x) ∨ GreenSignal(x)

C3: ¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,Ambulance).

Resolution 1: C1 + the two Ambulance facts → ClearPath(Ambulance).

Resolution 2: C2 + ClearPath(Ambulance) → GreenSignal(Ambulance).

Resolution 3: Instantiate C3 with x=CarA, y=Ambulance.

Resolution 4: Resolve the instantiated clause with Vehicle(CarA), Behind(CarA,Ambulance), and GreenSignal(Ambulance) → Proceed(CarA).

Hence Proceed(CarA) is logically entailed under the supplied rules.

Resolution Flowchart

Start → Emergency facts → Rule 1 → ClearPath(Ambulance) → Rule 2 → GreenSignal(Ambulance) → Behind(CarA,Ambulance) + Vehicle(CarA) → Rule 3 → Proceed(CarA) → End

(c) Two Simultaneous Emergency Vehicles

The present KB can reason about each emergency vehicle independently, but it does not provide an explicit mechanism for resolving conflicts when two emergency vehicles compete for the same route or intersection.

- Add facts identifying both emergency vehicles and their emergency status.
- Add a priority relation such as HigherPriority(x,y).
- Add route/intersection facts to identify conflicting paths.
- Add a rule preventing incompatible simultaneous green signals.
- Add a scheduling/conflict-resolution rule to select a safe order.

Therefore, additional priority, route and conflict-management knowledge is required for robust simultaneous-emergency handling. This is consistent with the assessment's request to identify additional rules or facts. filecite turn0file0 L43-L44

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

The KB contains three propositional rules and the facts SoilDry=True and CropWheat=True. filecite turn0file0 L45-L50

(a) CNF Conversion

Use A→B ≡ ¬A∨B and De Morgan's law where necessary.

P1: SoilDry → IrrigationNeeded ≡ ¬SoilDry ∨ IrrigationNeeded.

P2: (IrrigationNeeded ∧ CropWheat) → ApplyDripMethod ≡ ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod.

P3: (¬ApplyDripMethod ∧ CropWheat) → CropAtRisk ≡ ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk.

Facts: SoilDry and CropWheat are unit clauses.

Statement	CNF
P1	¬SoilDry ∨ IrrigationNeeded
P2	¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
P3	ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
Facts	SoilDry ; CropWheat

(b) Resolution Refutation for ApplyDripMethod

Goal: ApplyDripMethod.

Negate goal: ¬ApplyDripMethod.

Resolve P1 with SoilDry: IrrigationNeeded.

Resolve P2 with IrrigationNeeded: ¬CropWheat ∨ ApplyDripMethod.

Resolve with CropWheat: ApplyDripMethod.

Final resolution: ApplyDripMethod + ¬ApplyDripMethod → □.

Since the empty clause □ is obtained, the negated goal is inconsistent with the KB. Therefore ApplyDripMethod is proven.

Resolution tree:
(¬SoilDry∨IrrigationNeeded) + SoilDry → IrrigationNeeded
(¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod) + IrrigationNeeded → (¬CropWheat∨ApplyDripMethod)
(¬CropWheat∨ApplyDripMethod) + CropWheat → ApplyDripMethod
ApplyDripMethod + ¬ApplyDripMethod → □

(c) Removing SoilDry

Remaining fact: CropWheat.

P1: Cannot derive IrrigationNeeded because SoilDry is absent.

P2: Cannot derive ApplyDripMethod because IrrigationNeeded is absent.

P3: Cannot derive CropAtRisk because ¬ApplyDripMethod is not given.

Conclusion: CropAtRisk does not follow from the supplied KB.

Importantly, failure to prove ApplyDripMethod is not the same as proving ¬ApplyDripMethod. The assessment specifically asks for this distinction after removing SoilDry. filecite turn0file0 L55-L56

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

The supplied percepts are Stench at [1,2], Breeze at [1,1] and Glitter at [2,2]. □filecite□turn0file0□L58-L64□

(a) Belief State and Modus Ponens

Modus Ponens has the form A→B, A ⟹ B. Applying it to the supplied percept rules gives:

Percept	Inference	Conclusion
Stench[1,2]	Stench[1,2]→WumpusAdjacent[1,2]	Wumpus is adjacent
Breeze[1,1]	Breeze[1,1]→PitAdjacent[1,1]	Pit is adjacent
Glitter[2,2]	Glitter[2,2]→Gold[2,2]	Gold is at [2,2]
No negative hazard facts	Rule 4 needs ¬Wumpus and ¬Pit	Safety cannot be proven

(b) Forward Chaining

Step 1: Stench[1,2] → Wumpus is adjacent to [1,2].

Step 2: Breeze[1,1] → Pit is adjacent to [1,1].

Step 3: Glitter[2,2] → Gold is at [2,2].

Step 4: Rule 4 cannot fire because explicit ¬Wumpus and ¬Pit facts are not supplied.

Thus the KB identifies possible hazard adjacency and the gold location, but it does not uniquely identify exact Wumpus/Pit cells.

Forward-Chaining Flowchart

Start → Read percepts → Stench[1,2] → Wumpus adjacent → Breeze[1,1] → Pit adjacent → Glitter[2,2] → Gold[2,2] → Check negative hazard facts → Not available → Safety not proven → End

(c) Safety of Path [1,1] → [1,2] → [2,2]

[1,1]: Breeze is present, so a nearby Pit is possible; safety is not proven.

[1,2]: Stench is present, so a nearby Wumpus is possible; safety is not proven.

[2,2]: Glitter proves gold is present, but does not prove that the cell is hazard-free.

Overall: The complete path cannot be logically certified as safe from the supplied KB.

This conclusion follows the given rules rather than assuming additional standard Wumpus World rules. The assessment asks the path to be evaluated using only the knowledge derived in parts (a) and (b). □filecite□turn0file0□L67-L70□

COMPARATIVE ANALYSIS OF REASONING METHODS

Method	Direction	Purpose	Applied in
Forward Chaining	Facts → conclusions	Derive reachable facts	Cases 1 & 4
Backward Chaining	Goal → facts	Verify a selected goal	Case 1
Unification	Expressions → substitution	Match FOL terms	Case 2
Resolution	Clauses → contradiction	Prove entailment	Cases 2 & 3
CNF Conversion	Formula → clauses	Prepare for resolution	Case 3

Forward Chaining is data-driven and useful for deriving multiple conclusions. Backward Chaining is goal-driven and focuses on the rules that can prove a selected target. Unification allows variables in FOL rules to be matched with concrete entities. Resolution provides a systematic proof procedure, while CNF conversion makes formulas suitable for clause-based resolution.

KEY OBSERVATIONS

- All antecedent conditions of a rule must be satisfied before the rule can fire.
- Absence of a fact does not automatically establish its negation.
- Backward Chaining reduces a goal into smaller subgoals.
- An MGU should make expressions identical without unnecessary restrictions.
- Resolution refutation proves a goal by deriving the empty clause from the KB plus the negated goal.
- Incomplete knowledge bases can limit what an intelligent agent can safely conclude.

CONCLUSION

The four case studies demonstrate the role of logical reasoning in AI agents. The medical system uses propositional inference to derive possible diagnoses. The traffic system uses FOL, unification and resolution to reason about emergency routing. The agricultural system demonstrates CNF conversion and resolution refutation, including how removing a single fact can break an inference chain. The Wumpus World system shows how percepts can generate beliefs while also demonstrating that safety cannot be claimed without sufficient negative hazard information.

The most important principle across all four cases is that an AI agent should draw only conclusions supported by its knowledge base and inference rules. When information is missing, the correct logical response is to identify the limitation rather than inventing a fact. This makes the reasoning process transparent, explainable and suitable for intelligent decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand